

Object Oriented Programming Lab

1. Sign and Parity Check

- Input a number and print whether it is:
 - Positive Even
 - Positive Odd
 - Negative Even
 - Negative Odd
 - Zero

(Use nested if-else and conditional operator)

2. Largest and Smallest of Four Numbers

- Input 4 integers and determine both the **largest** and the **smallest** number.
(Students must use nested if-else or conditional operator chaining)

3. Electricity Bill Calculator (If-Else Ladder)

- Calculate bill as per units consumed:
 - First 100 units → ₹1.5/unit
 - Next 100 units → ₹2.5/unit
 - Beyond 200 units → ₹4.0/unit
- If bill > 500, add 10% surcharge.

4. Triangle Validator and Type Checker

- Input 3 sides of a triangle. Check:
 - If it forms a valid triangle.
 - If valid, determine whether it is **Equilateral, Isosceles, or Scalene**.
 - Also check if it is a **Right-angled Triangle**.

5. Student Grade with Division (If-Else + Nested Conditions)

- Input 5 subject marks, calculate percentage, and assign:
 - ≥ 90 → Grade A (Distinction)
 - 75–89 → Grade B (First Division)
 - 60–74 → Grade C (Second Division)
 - 40–59 → Grade D (Pass)
 - < 40 → Fail

6. Number to Words (Switch-Case)

- Input a digit (0–9) and print it in words.
Example: Input → 7, Output → "Seven".
- Extend the program to handle **two-digit numbers** (e.g., 42 → "Forty Two").

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7. Month and Days (Switch-Case + Leap Year)

- Input month number (1–12) and year.
- Print the **number of days** in that month, considering leap year rules for February.

8. Income Tax Calculator

- Input annual income and compute tax as per slabs:
 - $\leq 2,50,000 \rightarrow$ No tax
 - $2,50,001\text{--}5,00,000 \rightarrow 5\%$
 - $5,00,001\text{--}10,00,000 \rightarrow 20\%$
 - $10,00,000 \rightarrow 30\%$
- Display taxable income and tax amount.

9. Menu-Driven Mathematical Operations (Switch-Case)

- Provide options:
 - 1 $\rightarrow$ Factorial
 - 2 $\rightarrow$ Prime Number Check
 - 3 $\rightarrow$ Armstrong Number Check
 - 4 $\rightarrow$ Palindrome Number Check
 - 5 $\rightarrow$ Exit

(Students must implement logic for each case with proper conditions.)

10. Mini ATM Simulation (Switch + If-Else)

- Menu options:
 - 1 $\rightarrow$ Balance Inquiry
 - 2 $\rightarrow$ Cash Withdrawal (check balance before withdrawal)
 - 3 $\rightarrow$ Cash Deposit
 - 4 $\rightarrow$ Exit
- Maintain balance and update accordingly.